Q: Are all of these enough to get full marks in the exam?

A: NO. This is a practice sheet. Meaning, you can practice all you want using the questions from this sheet. However, doing well in exams depends upon your ability to understand a question, formulate an answer, and express it correctly. You see, these are humane skills that cannot be guaranteed by completing a practice sheet only. But yeah, Best of luck anyway.

Chapter 1 (Computer Abstractions and Technology)

Question - 1:

Assume a 19 cm diameter wafer, X has a cost of 20, contains 89 dies, and has 0.023 defects/cm^2. Assume another 20 cm diameter wafer, Y has a cost of 15, contains 100 dies, and has 0.031 defects/cm^2. Consider N = 2.

A. What do you understand by yield?
B. `Dies per wafer ≈ Wafer area / Die Area – Explain why there is “≈” not “=”`
C. Find the yield for both wafers.
D. Find the cost per die for both wafers.

Question - 2:

`Suppose gaming consoles PlayStation 5 and Xbox Series X use different implementations of AMD's Zen 2 architecture. The instructions they support can be divided into four classes according to their CPI (class A, B, C, and D). PlayStation has a clock rate of 2.7 GHz and the instruction classes have CPIs of 7, 2, 3, and 6 respectively whereas Xbox has a clock rate of 3.0 GHz and the instruction classes have CPIs of 5, 4, 2, and 1 respectively. Now suppose, a program has an instruction count of 1.0 * 10^6, and the instructions are`

divided into classes as follows:

30% class A,
50% class B,
10% class C, and
10% class D.

Now answer the following questions:

1. Calculate how many more clock cycles per instruction on average does the PlayStation take compared to the Xbox?
2. Calculate the difference between the execution time in these two consoles in milliseconds
3. If the program runs on a reference PC with an execution time of 120 ms, calculate the SPECRatio for the PlayStation

Question - 3:

Consider three different processors P1, P2, and P3 executing the exact same instruction set. P1 has a 3 GHz clock rate and a CPI of 1.5. P2 has a 2.5 GHz clock rate and a CPI of 1.0. P3 has a 4.0 GHz clock rate and has a CPI of 2.2.
Now answer the following questions:

a. Which processor has the highest performance expressed in instructions per second?
b. If the processors each execute a program in 10 seconds, find the number of cycles and the number of instructions.
c. We are trying to reduce the execution time by 30%, but this leads to an increase of 20% in the CPI. What clock rate should we have to get this time reduction?

Question - 4:

Consider a computer running a program that requires 250s, with 70s spent executing add instructions, 85s executed sub instructions, and 40s spent executing left-shift instructions.

a. By how much is the total time reduced if the time for add operations is reduced by 20%?
b. Can the total time can be reduced by 20% by reducing only the time for left-shift instructions?

Question - 5:

Processor, P1 has a clock rate of 4 GHz, average CPI of 0.9, and requires the execution of 5.0E9 instructions. Processor,P2 has a clock rate of 3 GHz, an average CPI of 0.75, and requires the execution of 1.0E9 instructions.

a. We consider the processor with the largest MIPS has the largest performance. See if this is true for Processor P1 and P2.

Question - 6:

Explain the power trend equation. If a new system has 14.3% less capacitive load and uses only 81.3% of the voltage and frequency of the old system, what percentage of power utilization can be reduced in the new system compared to the old system?

Question - 7

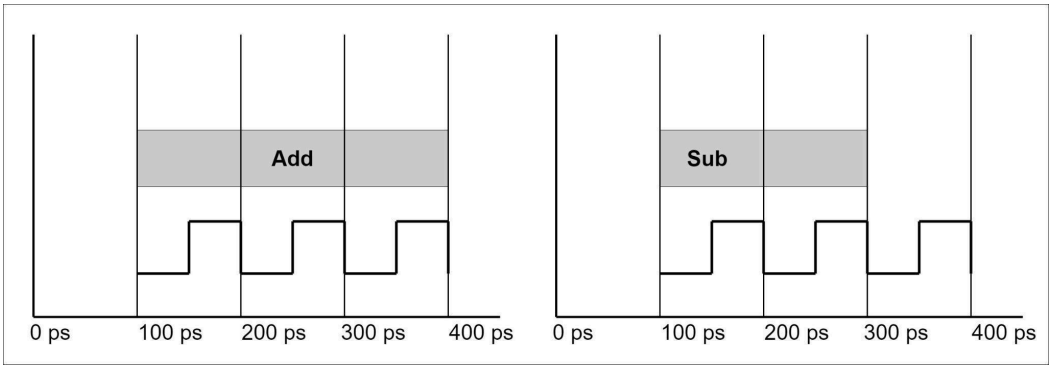


Figure 1: Represents 1 add instruction

Figure 2: Represents 1 sub instruction

Program **A** is divided into two classes according to their **CPI** (Add and Sub). The **instruction counts** are 21 and 3 respectively. Reference for **program A** is 1080ps.

Now, answer the following questions,

a) What is the Clock period? **Hint**: follow any of the figures
b) What is the frequency?
c) What is the CPI for Add and Sub ?
d) What is the Avg. CPI?

e) Find out the execution time of the program?
f) Find the SPEC ratio?
g) If you want to improve the performance by 1.2 times, what improvement do you need to include in the program's add operation?

Newly added problems:

[Solutions not given - hints added]

Question - 8

When summarizing the overall performance of a new processor across a suite of different programs (like the SPEC benchmark), computer architects calculate the performance ratio (speedup) for each program compared to a reference machine. To get the final, overall score, the industry strictly uses the **Geometric Mean** of these ratios.

Can we use the standard **Arithmetic Mean** (average) of these performance ratios? Why or why not? Explain with an example.

[*Hint:*

a. *Consider Computer A as the reference machine to calculate execution time ratios. Then calculate the overall score of the benchmark using Arithmetic mean.*
b. *Repeat the same calculations but this time consider Computer B as the reference machine.*

The answer of both steps will not agree when we use Arithmetic mean. What happens when we use Geometric mean to calculate the overall score?

Benchmark programs	*Computer A (Execution time)*	*Computer B (Execution time)*
Program 1	*10 s*	*100 s*
Program 2	*100 s*	*10 s*

]

Question - 9

For decades, processor performance scaled exponentially simply by increasing the clock frequency. In the mid-2000s, this trend abruptly stopped, and engineers shifted to building multicore microprocessors (putting more than one processor per chip). Looking at the CMOS power equation ($Power = Capacitive\,load \times Voltage^2 \times Frequency$), explain the improvement trend of processors over the generations.

[Hint:

Try to answer the following questions-

a. *Historically, how did engineers make a single processor finish its tasks faster? (Think about the relation of execution time and clock rate).*
b. *Why couldn't engineers increase clock speed indefinitely? (Think about the power equation)*
c. *Then how the engineers improved performance keeping power consumption the same? (Think about the voltage factor in the power equation)*
d. *Why couldn't they reduce voltage any further? What happened after that?**]***

Question - 10

Two friends, Alex and Ben, just finished building their ultimate high-end PCs. Alex built his using an Intel Core i9-14900K, while Ben built his using an AMD Ryzen 9 7950X.

Since they were immature and extremely competitive about their builds, they decided to settle the debate by running a massive data-sorting benchmark. Because their processors use different architectures, the compilers generated a different number of machine instructions for the exact same sorting task.

Here is the data from their benchmark run:

Alex's Intel PC: Executed 10×10^9 instructions and finished the benchmark in **5.0 seconds**.

Ben's AMD PC: Executed 15×10^9 instructions and finished the benchmark in **6.8 seconds**.

Ben quickly calculates his MIPS, looks at Alex's data, and starts bragging: *"My AMD processor is clearly superior to yours!"*

A third friend, who just finished studying Chapter 1 of Computer Architecture, overhears them and sighs. *"You guys are arguing over nothing. You absolutely cannot use MIPS as a valid performance metric to compare those two processors."*

a. Calculate the MIPS (Millions of Instructions Per Second) for both Alex's and Ben's processors. Based on your calculations, who has the higher MIPS?
   [***Hint:*** $\frac{Instruction\ Count}{Execution\ time \times 10^6}$***]***
b. Do you agree with the third friend? Explain exactly why MIPS can't identify the faster computer in this scenario.
   *[**Hint:** MIPS doesn't account for differences in ISA or complexity of instructions**]***

Question - 11

An imaginary processor is being evaluated using a simplified benchmark suite consisting of four distinct programs.

Program Name	Instruction Count ($\times 10^6$)	CPI	Clock Cycle Time (ns)	Reference Time (s)
GNU-Octave Comp.	2451	0.52	0.48	1.25
Robot AI Simulation	4521	1.23	0.48	3.14

Path Tracing	2347	0.95	0.48	1.75
x512 Compression	3416	1.06	0.48	2.06

Given the hardware and software metrics in the table above, perform the following tasks based on SPEC CPU benchmark evaluation standards:

a. Calculate the Execution Time (in seconds) for each program.

[Hint: $Execution\ time\ =\ IC\ \times CPI\ \times Clock\ time$***]***

b. Calculate the Performance Ratio (SPECratio) for each program.

[Hint: $Performance\ Ratio\ =\ \frac{Execution\ time}{Reference\ time}$***]***

c. Calculate the overall geometric mean of these performance ratios to determine the final summarized performance score of the new processor.

[Hint:

$$Performance\ score\ (geometric\ mean)\ =\ \sqrt[n]{\prod_{1}^{n} Performance\ ratio}$$***]***

Question - 12

You are optimizing the training pipeline for a new deep learning model. Profiling a single training epoch reveals the following execution time breakdown:

- Data Loading & Preprocessing (CPU): 25% of the total time
- Tensor Matrix Math (GPU): 60% of the total time
- Graph Evaluation & I/O: 15% of the total time

Your project advisor wants to reduce the total training time by a factor of 3 (an overall speedup of 3.0x). The advisor suggests taking the entire budget and buying a massive, state-of-the-art GPU cluster to accelerate *only* the Tensor Matrix Math.

a. Using Amdahl's Law, mathematically prove whether the advisor's goal of a 3.0x overall speedup is possible or impossible under this plan.

[Hint: $\frac{100}{3} = 25 + 15 + \frac{60}{n}$*]*

b. If you invest in a multi-core CPU upgrade that successfully makes the Data Loading & Preprocessing phase 5 times faster, exactly how much faster must you make the GPU Tensor Matrix Math to hit your advisor's strict 3.0x overall speedup target?

[Hint: $\frac{100}{3} = \frac{25}{5} + 15 + \frac{60}{n}$*]*

Question - 13

Processor A, released in 2010, had a clock rate of 4.0 GHz and a voltage of 1.3 V. Assume that, on average, it consumed 20 W of static power and 110 W of dynamic power.**Processor B**, released in 2018, has a clock rate of 3.0 GHz and a voltage of 0.8 V. Assume that, on average, it consumed 15 W of static power and 35 W of dynamic power.

a) For each processor, find the average capacitive load.

[Hint: Use the standard dynamic power equation P_dynamic = C * V^2 * f. Rearrange this equation to isolate and solve for the capacitance (C). To ensure the resulting unit is in Farads (F), make sure to convert the clock frequency from GHz to Hz by multiplying by 10^9.]

b) Find the percentage of the total dissipated power comprised by static power, and the ratio of static power to dynamic power for each technology.

[Hint: First, establish the total power by summing the static and dynamic power for each processor (P_total = P_static + P_dynamic). For the percentage, compute (P_static / P_total) * 100. For the ratio, simply divide the static power by the dynamic power (P_static / P_dynamic).]

c) If Processor A's total dissipated power is to be reduced by 15%, how much should the voltage be reduced to maintain the same leakage

current? *(Note: Power is defined as the product of voltage and current).*

*[Hint: Calculate your target total power, which is 85% of Processor A's original total power. Let x be the voltage reduction factor (V_new = x * V_old). Since the problem states leakage current remains constant, the new static power scales linearly with voltage (x * P_static). Since clock rate isn't mentioned as changing, assume it is constant, meaning dynamic power scales quadratically with voltage (x^2 * P_dynamic). Set up the quadratic equation: P_target = (x * P_static) + (x^2 * P_dynamic). Solve for the positive root of x, find V_new, and subtract it from the original 1.3 V to find the absolute reduction.]*

Question - 14

Assume for ALU, Memory (load/store), and Control (branch) instructions, a processor has CPIs of 2, 10, and 4, respectively. On a single processor, a baseline program requires the execution of $4.0 x 10^9 ALU instructions, 2.0 x 10^9 Memory instructions, and 500 x 10^6 Control instructions. Assume the processor operates at a 2.5 GHz clock frequency.

As the program is parallelized to run over multiple cores, the number of ALU and Memory instructions per processor is divided by 0.8 x p (where p is the number of processors, for p > 1). The number of Control instructions per processor remains constant regardless of p. For p = 1, use the baseline unparallelized instruction counts.

a) *Find the total execution time for this program on 1, 2, 4, and 8 processors, and show the relative speedup of the 2, 4, and 8 processors relative to the single processor.*

*[Hint: Execution time (T) is calculated using the CPU time formula: T = (Total Clock Cycles) / Clock Rate. First, determine the instruction count for each category based on the number of processors (p). For p > 1, apply the scaling factor (divide by 0.8 * p) to the ALU and Memory counts, while keeping the Control count constant. Calculate*

the total clock cycles by multiplying each instruction count by its CPI and summing the results. Divide by the 2.5 GHz clock rate to find the execution time. Finally, calculate the relative speedup using the formula: Speedup = T_1 / T_p.]

b) *If the CPI of the ALU instructions is doubled, what is the impact on the execution time of the program on 1, 2, 4, and 8 processors?*

[Hint: Change the CPI for ALU instructions from 2 to 4. Repeat the exact same total clock cycle and execution time calculations from the previous step for p = 1, 2, 4, and 8 using this new CPI value. Compare these new execution times to the original execution times to demonstrate the impact (e.g., state the new times and the absolute difference or percentage increase).]

c) *To what value must the CPI of Memory instructions be reduced for a single processor to exactly match the performance of four processors using the original CPI values?*

[Hint: Identify your target execution time, which is the execution time for 4 processors (T_4) calculated in the first question. Set up the execution time equation for a single processor (p = 1) using the baseline instruction counts, the original ALU and Control CPIs, but leave the Memory CPI as an unknown variable (x). Set this entire equation equal to T_4 and algebraically solve for x.]

Question - 15

The results of the SPEC CPU2006 bzip2 benchmark running on an AMD Barcelona has an instruction count of 2.389E12, an execution time of 750 s, and a reference time of 9650 s. Suppose that we are developing a new version of the AMD Barcelona processor with a 4 GHz clock rate. We have added some additional instructions to the instruction set in such a way that the number of instructions has been reduced by 15%. The execution time is reduced to 700 s and the new SPECratio is 13.7. Find the new CPI.

*[Hint: Start by calculating the new Instruction Count (IC) by applying the 15% reduction to the original baseline (New IC = 2.389E12 * 0.85). Next, identify the*

*variables needed for the CPU performance equation. You are given the new Execution Time (T = 700 s) and the new Clock Rate (f = 4 * 10^9 Hz). Note that the reference time and new SPECratio are distractor values included here—they aren't required to find the CPI since the actual execution time is explicitly provided. Finally, take the standard execution time formula, T = (IC * CPI) / f, and rearrange it to isolate your target variable: CPI = (T * f) / New IC.]*